THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
3	**(b)** the theory is strongly supported but remains open to tests	Science is provisional.
4	**(d)** A is false but R is true.	A is false (theories can be refined); R is true.
5	**(a)** 5.4 cm	Subtract the zero error.
6	**(d)** dosage of a medicine for a child	Safety critical.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(b)** 1 per cent	0.2/20.4.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Some readings are slightly too high and some too low, so the mean reduces the effect of random errors.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the SI unit does not explain why the least count is 1 mm (it is the smallest division on the scale).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area = 96 square cm (1); About 2 per cent (1).	1+1
12	A straight line shows proportionality (1); Easy to read the slope and spot errors (1).	1+1
13	Distance = 3 x 10^5 x 3.15 x 10^7 (1); = 9.45 x 10^12 km (1).	1+1
14	Mean = 9.76 m/s^2 (1); Range = 0.3 m/s^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1).	1+1+1
16	Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1).	1+1+1
17	Rising curve (1); Not a straight line (1); Longer pendulum has longer period (1).	1+1+1
	OR: Mean = 54.4 s for 30 (1); T = 1.81 s (1); Spread about 0.4 / 54.4 = 0.7 per cent (1).	1+1+1
18	Volume = 144 cubic m (1); Mass = 144 x 1.2 (1); = about 173 kg (1).	1+1+1
	OR: Informed consent (1); Safety tests first (1); Honest reporting of side effects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1 mm = 0.1 cm (1). (ii) $(14.2 + 14.4 + 14.3 + 14.1 + 14.5)/5 = 71.5/5 = 14.3$ cm (1). (iii) $14.3\ \text{cm} = 0.143$ m (1). (iv) Readings that are too high and too low cancel partly in the mean, so random errors are reduced (1); the largest deviation is $\lvert 14.5 - 14.3\rvert = \lvert 14.1 - 14.3\rvert = 0.2$ cm (1).	1+1+1+2
20	Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1).	1+1+1+1+1
	OR: Observations of planets (1); Copernicus's model (1); Galileo's telescope (1); Kepler's laws (1); Debate and acceptance (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying by $18/5$ instead of $5/18$ (or the reverse) when converting between km/h and m/s; remember that m/s values are always *smaller* numbers.
- Writing a measured value without its unit, or mixing units (cm with m) in one calculation.
- Confusing *law* with *theory*: a law describes a pattern, a theory explains it; a theory does not 'become' a law.

- Taking the mean of the readings as 'the true value'; it is only a more reliable estimate, and a single reading is not 'wrong' just because it differs from others.
- Treating a model as an exact copy of reality, or saying that a model that fails a test should be ignored instead of being improved.
- Mixing up accuracy and precision, and reading the least count of an instrument wrongly (use the smallest division, not the numbered one).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15 per cent	3 x 5 per cent.
2	**(b)** to test the procedure and find problems	Trial run.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 4.5 x 10^-4	Standard form.
5	**(b)** precise but not accurate	Close to each other but away from true.
6	**(d)** A is false but R is true.	A false; R true.
7	**(d)** 0.5 m	Uncertainty must exceed the difference.
8	**(a)** accurate on average but not precise	Mean close to the true value.
9	**(c)** A is true but R is false.	Models keep only the features that matter for the question, so they are not exact copies; R is false.
10	**(d)** A is false but R is true.	A is false because models are revised when evidence demands it; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Millions of data points and equations (1); Many calculations are needed quickly (1).	1+1
12	Total energy must remain constant (1); A violation signals an error in the model (1).	1+1
13	A future observation may contradict it (1); Science always remains open to revision (1).	1+1
14	Zero error shifts all readings in the same direction (1); Reaction time varies unpredictably (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Must be possible to show it false (1); Example of testable: heating increases the rate of dissolving (1); Not falsifiable: unseen forces cause everything (1).	1+1+1
16	1 litre = 1000 mL (1); 1000/0.05 (1); = 20000 drops (1).	1+1+1
17	May be due to chance or error (1); Reproducibility is the test of reliability (1); Needs further study (1).	1+1+1
	OR: Error = 0.02 (1); 0.02/2.70 x 100 = 0.74 per cent (1); Careful measurement of mass and volume (1).	1+1+1
18	Units of distance x time are m s (1); Speed has unit m/s (1); Unit mismatch shows error (1).	1+1+1
	OR: Melting point depends slightly on pressure (1); Condition must be clear for repeating (1); Conditions are part of a law (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $240/4 = 60$ km/h (1). (ii) $60 \times \frac{5}{18} = \frac{50}{3} \approx 16.7$ m/s (1). (iii) $240\ \text{km} = 240 \times 1000 = 2.4 \times 10^5$ m (1). (iv) $25 \times \frac{18}{5} = 90$ km/h (1); the second bus is faster by $90 - 60 = 30$ km/h (1).	1+1+1+2
20	Standard units help sharing data (1); Mars Climate Orbiter used mixed units (1); Spacecraft was lost (1); Importance of checking units (1); Open data and cooperation (1).	1+1+1+1+1
	OR: Same mass of water and same container (1); Same mass of each fuel burnt (1); Same starting temperature and distance of flame (1); Repeat and average (1); Errors: heat loss, draught, reading errors (1).	1+1+1+1+1

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- Treating a model as an exact copy of reality, or saying that a model that fails a test should be ignored instead of being improved.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the gas is at low pressure	Large separation.
2	**(d)** A is false but R is true.	A is false because a hypothesis is a proposed explanation; R is true.
3	**(d)** 1.08 x 10^9	3 x 10^5 x 3600 km/h.
4	**(b)** 25000	2.5 x 10000.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 0.3 per cent	0.03/9.81 x 100.
7	**(d)** 12.34 g	Matches least count.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Ignoring friction simplifies the situation; R is the reason idealisations are used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the newton is a derived unit and does not explain why the first three are base units.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Theories are well-tested explanations (1); They are not just guesses (1).	1+1
12	5.4 x 10^4 x 1000 / 3600 (1); = 15000 m/s (1).	1+1
13	Experts check methods and conclusions (1); Mistakes are corrected before acceptance (1).	1+1
14	Two things occurring together need not be linked as cause and effect (1); e.g. ice cream sales and drowning rise in summer (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Density = 2.7 g/cm3 (1); = 2700 kg/m3 (1); Aluminium (1).	1+1+1
16	Tests the method on a small scale (1); Finds problems early (1); e.g. trial run of a dissolving experiment (1).	1+1+1
17	Compare methods and calibration (1); Repeat with the same procedure (1); Seek independent confirmation (1).	1+1+1
	OR: Control plants without tonic (1); Same soil and water (1); Compare growth (1).	1+1+1
18	Small-angle assumption breaks down (1); Period depends on amplitude then (1); Hence the model needs revision (1).	1+1+1
	OR: Saha: stellar ionisation (1); Raman: scattering of light (1); Their work used careful experiments and reasoning (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A model is a simplified representation of an object or system that keeps the features important for the question being studied (1). (ii) Atoms are not hard coloured balls and bonds are not rigid sticks, so it cannot show the real size, motion or electron arrangement (1). (iii) Any one: the solar-system model of the atom, a map, a globe, a particle model of matter (1). (iv) They check the measurements and assumptions and then modify or replace the model (1), because a model is accepted only as long as it agrees with evidence (1).	1+1+1+2
20	Check other factors like irrigation (1); Control experiments (1); Statistical analysis (1); Replicate across regions (1); Correlation does not prove causation (1).	1+1+1+1+1
	OR: Speed = distance/time (1); Direct: distance vs time at constant speed (1); Inverse: time vs speed at fixed distance (1); Graphs show relations (1); Units check formulas (1).	1+1+1+1+1

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